Alexander Pugantsov

Padua, Veneto, Italy – Open to Remote/EOR (EU)
Email: **a dot pugantsov at icloud dot com**
URL: https://pugantsov.com

Research Profile

Applied ML Scientist focused on evaluating and improving ML systems under deployment constraints, with expertise in fairness, privacy, and retrieval/ranking and classification. I design reproducible benchmarks and large-scale experiments, quantify trade-offs and risks (e.g., bias, privacy leakage), and provide decision-ready evaluation for model selection and deployment.

Research Experience & Education

2024 – **Postdoctoral Researcher, University of Padua** — Padua, Veneto, Italy
Privacy-preserving fairness and diversity estimation.

- Built a reproducible benchmark for evaluating demographic and fairness estimators under realistic deployment constraints.
- Led large-scale experiments across classification, text ranking, and vision tasks, analyzing robustness, distribution shift, and uncertainty.
- Evaluated privacy risks in aggregate estimation systems by constructing adversarial query strategies that bypass common output-level defenses, quantifying trade-offs between accuracy and formal privacy guarantees.

2019 – 2024 **Ph.D., University of Glasgow** – Glasgow, United Kingdom
Effective intermediate task selection in transfer learning.

- Developed an end-to-end framework for predicting effective transfer learning task combinations without exhaustive retraining.
- Designed structured dataset representations to characterize task distributions and applied a suite of divergence measures to estimate task similarity and predict relative transfer gain.
- Reduced training cost by up to 96.5% while maintaining downstream performance, improving efficiency and sustainability of transfer workflows.

2014 – 2019 **M.Sci., Software Engineering, University of Glasgow**

Deep learning & user modelling in financial social media

- Built large-scale recommender and classification systems over 17M+ financial social media posts.
- Implemented collaborative filtering, factorisation machines, sequential LSTM/CNN models, and attention-based networks for ranking and prediction tasks.
- Enriched sparse metadata using Wikidata/DBPedia via SPARQL and trained a hierarchical attention network achieving 78% classification accuracy.

Industry Experience

2023 – 2024 **Machine Learning Engineer, NDA** – Contract/Freelance

- Improved a semantic retrieval pipeline by training cross-encoder rerankers and contrastive embedding models, outperforming a hosted embedding baseline.
- Built a large ontology from noisy hierarchical data using LLM-assisted enrichment and graph-based topic modeling (community detection + topic refinement) to improve clustering and navigation.
- Implemented acronym disambiguation for domain-specific corpora using PEFT/LoRA with quantized LLMs, producing an acronym lexicon for query expansion in short, acronym-heavy documents.

2019 **Automation Engineer, Barclays** - Glasgow, United Kingdom

- Built internal automation to streamline screening workflows, reducing manual review effort and improving operational throughput for compliance teams.

2018 **Software Engineering Intern, Scoop Markets** - Glasgow, United Kingdom

- Implemented a Go-based aggregation pipeline for recommender outputs, surfacing model signals to improve monitoring and interpretability of entity-level predictions.

Technical Skills

Core: Python; PyTorch; TensorFlow; scikit-learn; Hugging Face.

Focus: large-scale evaluation/benchmarking; fairness & privacy; robustness under distribution shift; uncertainty analysis.

Engineering: Git; Docker; Linux; CI/CD.

Languages: English (Native), Russian (Native), Italian (Beginner)

Selected Publications

[C2] A. Pugantsov, R. McCreadie: Divergence-based domain transferability for zero-shot classification. *European Chapter of the Association for Computational Linguistics.* 2023. DOI: 10.18653/v1/2023.findings-eacl.122.

[C1] A. Pugantsov, R. McCreadie: Identifying suitable tasks for inductive transfer through the analysis of feature attributions. *European Conference on Information Retrieval.* 2022. DOI: https://doi.org/10.1007/978-3-030-99739-7_16